

Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination, Jan -2025

Applications of AI in Medicine [CSE-4014]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss the machine learning and data mining in terms of their goals, techniques, and applications. (5)
 - A)
 - B) Discuss any four types of distance measures used in K means clustering. (5)
- 2) Demonstrate the Total Dermatoscopic Score (TDS) formula and discuss the melanoma diagnosis with threshold values. (5)
 - A)
 - B) Demonstrate with a neat diagram the steps involved in training through FF-ANN. (5)
- 3) Discuss the following multiclass methods: (5)
 - (i) OvsA
 - (ii) Minimal Output Coding
 - A)
 - B) Explain the working of the Naïve Bayesian classifier. (5)
- 4) Discuss briefly the pap smear cervical screening and diagnostic techniques. (5)
 - A)
 - B) Explain briefly about clinical decision support. (5)
- 5) Discuss the advantages, disadvantages, and applications of NLP. (5)
 - A)
 - B) Design and explain the architecture of the intelligent personal health record system. (5)

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